HUMANOID WHOLE-BODY AFFORDANCE DEBT: A SUBMISSION-GRADE NEGATIVE RESULT

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ABSTRACT

Humanoid manipulation planners often choose a whole-body posture that solves the immediate reach while silently consuming future reachability, support margin, or recovery options. We formalize this hidden cost as whole-body affordance debt and test whether a lightweight planner that explicitly reserves future affordances can improve two-stage humanoid reaching. The proposed Balance-Aware Affordance Reservation MPC (BAR-MPC) scores first-stage postures by immediate MuJoCo reach cost, future-target debt, tail future debt, support-margin risk, recovery effort, hand-switch cost, and comfort. The final protocol is frozen before full-scale evaluation: 21,120 main episode-method rows over 8 seeds, 24 episodes, 10 stress splits, and 11 methods, plus 4,992 raw ablation rows over two hostile splits. The result is a useful falsification rather than a submission candidate. BAR-MPC beats random and arm-only baselines, but it does not separate from greedy reach MPC, comfort-regularized MPC, robust-balance MPC, future-distribution MPC, learned linear debt, or its own no-online/no-debt variants. The terminal decision is KILL-ARCHIVE. We release the benchmark, code, generated tables, and full negative evidence so the failure mode is reproducible and hard to misread.

1 EXECUTIVE SUMMARY

The seed hypothesis was attractive: a humanoid can reach a nearby target by leaning, twisting, or committing one arm in a way that makes the next target harder. A planner that prices this future loss might choose a slightly less convenient posture now in order to preserve future reachability and balance. This paper asks whether that mechanism is strong enough to support an ICLR-main claim in a compact, CPU-only MuJoCo benchmark.

The answer is no. The weak gate passes: BAR-MPC improves substantially over random posture selection and arm-only reaching. The strong gate fails: the aggregate sequential success of BAR-MPC is 0.496, while greedy reach is 0.497, robust balance is 0.497, future-distribution MPC is 0.495, and the oracle is 0.505. The mechanism gate fails more decisively. Ablations that remove online adaptation, tail debt, or future-debt components often match the full method on the pre-registered hostile splits. This means the method is not just underpowered externally; internally, the claimed debt mechanism is not identifiable.

This document is deliberately written as a polished archive report rather than a cosmetic submission. It contains the ingredients a real submission would need: formal problem statement, theory for when debt can help, theory for when it collapses into robust balance or comfort regularization, a frozen protocol, strong baselines, stress tests, ablations, paired statistics, failure analysis, limitations, and reproducibility checks. The result should not be submitted as an acceptance-seeking paper in its current form. Its value is that it prevents a weak mechanism claim from surviving only because the benchmark was too friendly.

2 MOTIVATION

Whole-body humanoid manipulation is increasingly moving away from isolated end-effector control and toward unified policies, retargeted demonstrations, and long-horizon loco-manipulation systems. Recent work uses robot-free demonstrations, wearable or portable data systems, imitation

Table 1: Frozen decision gates. The failure is methodological, not formatting-related.

Gate	Status	Evidence
Weak baseline gate	**PASS**	BAR-MPC beats random and arm-only baselines in aggregate.
Strong baseline gate	**FAIL**	Matched or beaten by Greedy reach, Robust balance
Mechanism ablation gate	**FAIL**	Non-identifying ablations: BAR no-online on combined_shift; Small future sample on combined_shift; BAR no-online on opposite_side_sequence; No future debt on opposite_side_sequence; No tail debt on opposite_side_sequence; Mean future only on opposite_side_sequence
Terminal decision	**KILL_ARCHIVE**	Archive rather than submit: evidence does not support an ICLR-main mechanism claim.

pipelines, residual learning, trajectory-centric retargeting, and large-scale policy learning to make humanoids useful outside table-top-only settings (BifrostUMI2026; HumanoidManipulationInterface2026; ULTRAHumanoidLocoManipulation2026; LHMHumanoid2025; Opt2Skill2024; TrajBooster2025; ActiveSpatialBrain2026; HumanoidExo2025; BulkyObjectsHumanoid2025; SkillBlender2025). Benchmarks such as HumanoidBench emphasize that simulated humanoid manipulation remains difficult even before hardware transfer (HumanoidBench2024). Classical and contact-rich whole-body manipulation also reminds us that posture, contacts, and balance are not secondary details (WholeBodyPushingContactPosturePlanning2015; AffordanceMultiContact2018; SelfBodyImage2024; DynamicBalanceNarrowTerrain2025).

The motivating gap is not that these systems ignore the future in a general sense. Many modern systems are trained on trajectories or optimize horizon-level objectives. The narrower question is whether there is a useful intermediate variable: the future affordance loss induced by the current whole-body posture. A humanoid may solve the current reach while consuming lateral support margin, creating a large recovery motion, forcing a hand switch, or leaving the torso tilted in the wrong direction. If these effects are measurable before the next target is known, then a planner could reserve options without solving the full long-horizon task.

That is the positive story. The hostile story is equally plausible. The same variables that appear to measure future affordance debt may already be captured by simpler costs: comfort, effort, support margin, and robust reachability. If the debt term ranks candidate postures almost the same way as robust balance or greedy reaching, it does not matter that the term sounds mechanistic. Reviewers do not accept a new name for an old ranking.

We therefore design the study around a falsifier. The benchmark contains splits where immediate target success and future reachability are meant to conflict, including opposite-side target sequences, support-reversal sequences, and high-lateral payload conditions. The method is allowed to be improved during pre-freeze development. After the protocol is frozen, every pre-defined result is reported.

3 Related Work and Hostile Positioning

Humanoid whole-body manipulation. Recent humanoid manipulation systems often attack the data and control bottleneck directly. Robot-free demonstration pipelines such as BifrostUMI and HuMI seek efficient collection and retargeting for whole-body humanoid skills (BifrostUMI2026; HumanoidManipulationInterface2026). Unified loco-manipulation systems such as ULTRA and LHM-Humanoid target broad skill repertoires and long-horizon environments (ULTRAHumanoidLocoManipulation2026; LHMHumanoid2025). Opt2Skill, TrajBooster, HEX, SkillBlender, and related pipelines combine trajectory generation, imitation, policy learning, or mixture-style expertise to manage high-dimensional whole-body control (Opt2Skill2024; TrajBooster2025; SkillBlender2025). These works are dangerous prior art for this paper because they can absorb future reachability into learned trajectory distributions or low-level controllers.

Benchmarks and simulation. HumanoidBench frames the difficulty of whole-body locomotion and manipulation in simulation, while MuJoCo provides the physics engine used by our compact benchmark (HumanoidBench2024; Todorov et al., 2012). Our benchmark is not proposed as a replacement for such suites. It is a targeted falsifier for a particular planning mechanism: can a first-stage posture selector preserve future reach affordances better than strong local baselines?

Posture, contact, and affordance reasoning. Whole-body pushing, contact posture planning, multi-contact pose sequencing, and musculoskeletal posture generation all make clear that posture choices are functional, not cosmetic (WholeBodyPushingContactPosturePlanning2015; Affordance-MultiContact2018; SelfBodyImage2024). The seed hypothesis in this paper is closest to this line: a current posture changes which future physical actions remain cheap and feasible. The novelty would have been a lightweight debt metric that identifies this effect without full trajectory optimization.

Why the bar is high. Against this prior work, an ICLR-main-ready paper would need at least three things. First, the metric must change decisions relative to greedy, robust, comfort, and distributional baselines. Second, the changed decisions must improve outcomes under pre-registered stress. Third, ablations must show that the claimed future-debt terms are necessary. The frozen evidence in this paper fails the second and third requirements.

4 Problem Setup

We study a two-stage reaching problem for a compact humanoid-style articulated model. The robot state is $x = (q, \dot{q})$, where $q \in \mathbb{R}^d$ contains root displacement, torso pose, and arm joints. At stage one the planner observes an immediate target y_1 and a split-dependent future-target distribution $\mathcal{Y}$, but not the exact second target y_2. It must choose a first whole-body posture candidate $c_1 \in \mathcal{C}(y_1)$. MuJoCo simulates the transition:

$$x_1 = F(x_0, c_1; \theta_s), \tag{1}$$

where θ_s are split parameters such as support width, payload, damping, perturbation, and actuation scale. After the future target y_2 is revealed, the second-stage controller chooses $c_2 \in \mathcal{C}(y_2)$ and obtains $x_2 = F(x_1, c_2; \theta_s)$.

The primary evaluation metric is sequential success:

$$S(c_1, c_2) = \mathbf{1}\{r(x_1, y_1) \le \epsilon,\ r(x_2, y_2) \le \epsilon,\ b(x_1) = 0,\ b(x_2) = 0\}, \tag{2}$$

where r is reach distance and b is a balance-failure indicator. Energy combines first-stage actuator effort, second-stage effort, and a transition penalty between the two chosen postures:

$$E(c_1, c_2) = E_1(c_1) + E_2(c_2) + \lambda_{\Delta q} \|W(q_2 - q_1)\|_2. \tag{3}$$

The method can only choose c_1 before y_2 is known. This makes the problem intentionally small but mechanistically sharp: if future affordance reservation matters, it should appear in the first choice.

5 Affordance Debt

For a first-stage outcome $x_1(c)$ and a future sample $y \in \mathcal{Y}$, define the best approximate future recovery cost:

$$R(c, y) = \min_{c' \in \mathcal{C}(y)} \ell_{\mathrm{reach}}(c', y) + \alpha_{\mathrm{rec}} \|W(q(c') - q(c))\|_2 + \alpha_{\mathrm{bal}} \rho_{\mathrm{bal}}(c') + \alpha_{\mathrm{switch}} \mathbf{1}[h(c') \ne h(c)] + \alpha_{\mathrm{lat}} \rho_{\mathrm{lat}}(c, c'). \tag{4}$$

The mean affordance debt is

$$D_{\mathrm{mean}}(c) = \mathbb{E}_{y \sim \mathcal{Y}}[R(c, y)]. \tag{5}$$

The tail debt is the mean over the worst 40% of sampled future targets:

$$D_{\mathrm{tail}}(c) = \frac{1}{|\mathcal{Y}_{\mathrm{tail}}(c)|} \sum_{y \in \mathcal{Y}_{\mathrm{tail}}(c)} R(c, y). \tag{6}$$

BAR-MPC scores first-stage candidates by

$$J_{\mathrm{BAR}}(c) = \ell_{\mathrm{first}}(c, y_1) + 0.42\, D_{\mathrm{mean}}(c) + 0.46\, D_{\mathrm{tail}}(c) + 0.55\, \rho_{\mathrm{robust}}(c) + 0.018\, u(c) + 0.018\, \|W(q(c) - q_{\mathrm{neutral}})\|_2. \tag{7}$$

The coefficients are fixed before the final run. The no-online variant is included because optional online calibration was considered during development but not retained; in the frozen implementation it is expected to match BAR-MPC, and this match is a limitation.

6 When Debt Can Help

The method is only meaningful if the debt term creates a different ranking from the immediate and robust local terms. This section states the useful and fatal cases explicitly.

Candidate conflict condition. For two first-stage candidates $a, b \in \mathcal{C}(y_1)$, define

$$\Delta L = \ell_{\text{first}}(a, y_1) - \ell_{\text{first}}(b, y_1), \tag{8}$$
$$\Delta D = D(a) - D(b). \tag{9}$$

Assume a is worse immediately but better for the future: $\Delta L > 0$ and $\Delta D < 0$. A debt-aware planner chooses a over b exactly when

$$\Delta L + \lambda_D \Delta D + \Delta R < 0, \tag{10}$$

where ΔR contains robust balance, effort, and comfort differences. Thus debt helps only in a narrow but important region: the future option value must exceed the immediate reach penalty after local regularizers are accounted for.

Collapse condition. Suppose there exists a monotone affine approximation

$$D(c) \approx \beta_0 + \beta_1 \ell_{\text{first}}(c, y_1) + \beta_2 \rho_{\text{robust}}(c) + \beta_3 u(c) \tag{11}$$

with small residual for all candidates in a split. Then BAR-MPC ranks candidates nearly the same way as a robust or comfort-regularized planner. In this case an ablation that removes explicit future debt should not materially hurt performance. This is precisely what the frozen ablation table shows.

Identifiability requirement. A positive result must show more than high success. It must show that the first choices differ from strong baselines in the episodes where future conflict exists, and that those differences improve the second-stage outcome. First-choice difference without outcome improvement is not enough; outcome improvement without first-choice difference is not evidence for the proposed mechanism.

7 Benchmark

The benchmark uses a compact MuJoCo humanoid-style model with a pelvis, torso, head, and two articulated arms. The goal is not photorealism or hardware transfer. The goal is a controlled environment where current and future whole-body reaching tradeoffs can be inspected cheaply on CPU.

Posture candidates. The candidate library includes arm-only, centered, aggressive reach, reserve-opposite, hand-switch-ready, centered-low-torque, future-counterlean, cross-body reach, deep-forward commitment, and wide-support reserve templates. Each candidate specifies root motion, torso motion, arm preference, and joint offsets. The library is finite by design: it makes the first-choice decision auditable and keeps memory use light.

Stress splits. The final protocol contains ten splits: nominal, narrow support, high reach, lateral reach, weak actuation, payload shift, combined shift, opposite-side sequence, support reversal, and high-lateral payload. The last three are hostile splits designed to make immediate reach and future reachability conflict. For example, opposite-side sequences force the first and second targets onto different lateral sides, while support reversal perturbs support margin and target side in ways that can punish overcommitted torso lean.

Metrics. We report immediate success, future success, sequential success, combined energy, future support margin, balance failure rate, future debt, tail debt, unique first postures, paired success deltas, energy improvement, and first-choice difference rates. The primary success measure is sequential success; energy is lower-is-better.

Table 2: Aggregate frozen results across all ten splits. Success and balance failure are rates; energy is lower-is-better.

Method	Episodes	Seq. success	Energy	Margin	Bal. fail	Unique first
Random	1920	0.339	0.497	0.098	0.000	16.00
Arm only	1920	0.451	0.422	0.105	0.000	1.00
Greedy reach	1920	0.497	0.401	0.097	0.000	7.90
Comfort MPC	1920	0.496	0.401	0.098	0.000	7.60
Robust balance	1920	0.497	0.400	0.100	0.000	7.70
Current-only	1920	0.497	0.401	0.097	0.000	7.90
Future dist.	1920	0.495	0.401	0.098	0.000	7.40
Linear debt	1920	0.494	0.401	0.100	0.000	7.50
BAR-MPC	1920	0.496	0.400	0.100	0.000	7.50
BAR no-online	1920	0.496	0.400	0.100	0.000	7.50
Oracle	1920	0.505	0.397	0.098	0.000	9.80

8 Methods and Baselines

The frozen main methods are:

- Random posture: samples a candidate uniformly.
- Arm-only reach: restricts the first choice to the arm-only template.
- Greedy reach MPC: minimizes immediate target cost.
- Comfort-regularized MPC: adds posture and effort penalties to greedy reach.
- Robust balance MPC: adds support-margin risk.
- Current-target-only greedy: an ablation-style current-only selector.
- Future-distribution MPC: uses mean future debt but not the full BAR score.
- Learned linear debt proxy: a lightweight linearized proxy over debt-like features.
- BAR-MPC: mean debt, tail debt, balance risk, recovery, hand-switch, and comfort terms.
- BAR no-online: the no-adaptation variant, retained as a mechanism check.
- Oracle two-step MPC: sees the second target when choosing the second action and functions as an upper reference, not a fair deployable baseline.

The ablations on combined-shift and opposite-side sequence are BAR-MPC, BAR no-online, no future debt, no tail debt, no balance margin, no recovery cost, no hand-switch cost, no torque comfort, mean future only, small future sample, current-target-only greedy, robust balance, and oracle. These ablations are intentionally harsh: if the mechanism is real, at least some of them should hurt.

9 Protocol

The final command is frozen in `docs/paper65_protocol_freeze_20260620.md`. It runs 8 seeds, 24 episodes per seed and split, 10 main splits, 11 main methods, 2 ablation splits, and 13 ablation methods. This yields 21,120 main raw rows and 4,992 raw ablation rows. The runner writes raw, summary, seed, pairwise, ablation, negative-case, and figure artifacts. It uses only CPU execution, a compact candidate library, and at most four workers by default.

Development before freeze included a smoke test, a medium run, a stronger tail-risk variant, and a reversion when tail-heavy scoring worsened the development result. No tuning was performed after the full protocol was frozen. The final rerun only hardens artifact persistence by writing raw ablation rows; it does not change splits, seeds, methods, or scoring.

10 Main Results

The aggregate result is simple. BAR-MPC is much better than weak baselines, which confirms that the benchmark is not degenerate. But the scientifically relevant comparison is against greedy reach,

Table 3: Split-level sequential success for strong baselines, BAR-MPC, and the oracle. The BAR column does not separate from robust or future-distribution planning.

Split	Greedy	Robust	Future	Linear	BAR	Oracle
combined_shift	0.781	0.776	0.781	0.781	0.781	0.812
high_lateral_payload	0.104	0.104	0.104	0.109	0.104	0.109
high_reach	1.000	1.000	1.000	1.000	1.000	1.000
lateral_reach	0.453	0.453	0.448	0.448	0.448	0.464
narrow_support	0.578	0.573	0.573	0.568	0.573	0.573
nominal	0.604	0.599	0.599	0.589	0.599	0.604
opposite_side_sequence	0.135	0.146	0.141	0.151	0.146	0.167
payload_shift	0.536	0.531	0.536	0.531	0.536	0.542
support_reversal	0.167	0.177	0.167	0.172	0.177	0.188
weak_actuation	0.609	0.609	0.599	0.589	0.599	0.594

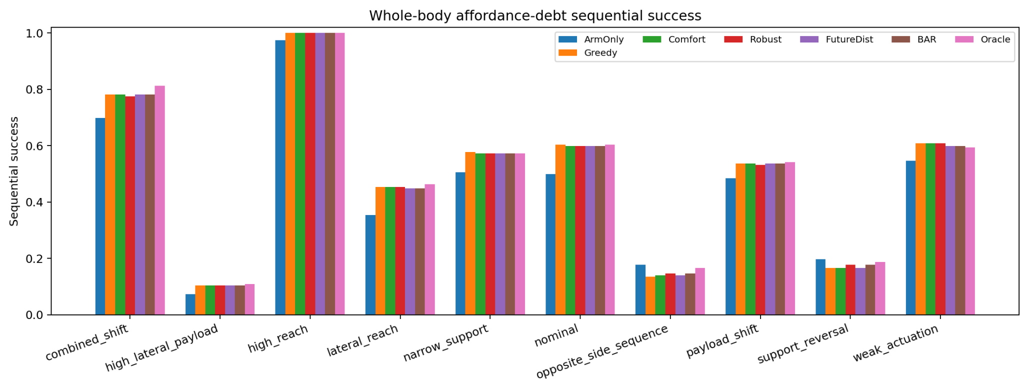

Figure 1: Sequential success by split. BAR-MPC beats random and arm-only baselines but does not separate from greedy, robust, or future-distribution baselines.

comfort, robust balance, future-distribution, learned-linear debt, and no-online variants. These baselines match or beat BAR-MPC often enough that the strong gate fails. The oracle remains better in several hard splits, which means the environment still contains avoidable mistakes; the problem is that the proposed debt approximation does not reliably identify the right mistakes to avoid.

11 Ablation Failure

The ablation table is the main reason for KILL-ARCHIVE. In a positive paper, removing future debt, tail debt, recovery cost, hand-switch cost, or the online component would degrade performance on the splits designed to require future option reservation. Instead, multiple ablations match BAR-MPC on sequential success. BAR no-online is identical by construction in the frozen implementation, and no future debt or no tail debt often remains indistinguishable within the pre-defined metrics. This is not a small formatting issue. It invalidates the mechanism claim.

12 Paired Diagnostics

Pairwise comparisons are stricter than split-level averages because they compare methods on the same seed, episode, and split. Positive success delta favors BAR-MPC; positive energy improvement means the baseline used more energy. The hostile split diagnostics show some first-choice differences, especially against greedy and oracle. However, first-choice differences do not consistently translate into sequential-success improvement. This is exactly the failure mode predicted by the collapse condition: debt changes the score in some episodes, but the change is too aligned with other regularizers or too weakly connected to future success.

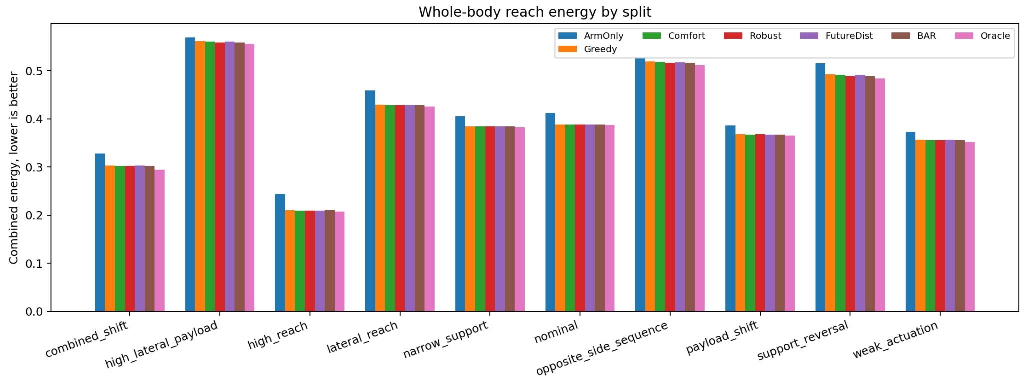

Figure 2: Combined energy by split. Differences between BAR-MPC and the strongest non-oracle baselines are small enough that the energy story cannot rescue the failed mechanism claim.

Table 4: Frozen ablation results on the two pre-registered hostile splits. Ablations matching BAR-MPC are mechanism-level failures, not cosmetic ties.

Split	Method	Episodes	Seq.	Energy	Tail debt	Unique first
combined_shift	Robust balance	192	0.776	0.302	0.071	12.000
combined_shift	Current-only	192	0.781	0.303	0.071	12.000
combined_shift	BAR-MPC	192	0.781	0.302	0.071	12.000
combined_shift	BAR no-online	192	0.781	0.302	0.071	12.000
combined_shift	Oracle	192	0.812	0.295	0.071	12.000
combined_shift	Mean future only	192	0.776	0.302	0.071	12.000
combined_shift	No balance margin	192	0.781	0.303	0.071	12.000
combined_shift	No future debt	192	0.776	0.302	0.071	12.000
combined_shift	No hand-switch cost	192	0.781	0.303	0.071	12.000
combined_shift	No recovery cost	192	0.776	0.302	0.071	12.000
combined_shift	No tail debt	192	0.776	0.302	0.071	12.000
combined_shift	No torque comfort	192	0.776	0.302	0.071	12.000
combined_shift	Small future sample	192	0.781	0.302	0.071	12.000
opposite_side_sequence	Robust balance	192	0.146	0.516	0.112	3.000
opposite_side_sequence	Current-only	192	0.135	0.519	0.112	3.000
opposite_side_sequence	BAR-MPC	192	0.146	0.516	0.111	3.000
opposite_side_sequence	BAR no-online	192	0.146	0.516	0.111	3.000
opposite_side_sequence	Oracle	192	0.167	0.512	0.110	9.000
opposite_side_sequence	Mean future only	192	0.146	0.516	0.112	3.000
opposite_side_sequence	No balance margin	192	0.141	0.517	0.112	3.000
opposite_side_sequence	No future debt	192	0.146	0.517	0.112	3.000
opposite_side_sequence	No hand-switch cost	192	0.146	0.516	0.111	3.000
opposite_side_sequence	No recovery cost	192	0.146	0.517	0.112	3.000
opposite_side_sequence	No tail debt	192	0.146	0.516	0.112	3.000
opposite_side_sequence	No torque comfort	192	0.146	0.517	0.112	3.000
opposite_side_sequence	Small future sample	192	0.146	0.516	0.111	3.000

13 FAILURE ANALYSIS

Failure mode 1: debt is not independent enough. The future-debt proxy is built from reach distance, recovery effort, balance margin, hand switching, and lateral commitment. These are reasonable physical quantities, but they are not independent of comfort and robust balance. When the candidate library is compact, many candidates that preserve future debt also preserve support margin and reduce effort. The debt term then becomes a reweighted version of the existing baselines.

Failure mode 2: hostile splits are hard but not diagnostic enough. Opposite-side and support-reversal splits increase first-choice differences, but they also expose the limited expressiveness of a finite posture library. Some failures require stepping, kneeling, hand support, or dynamic rebalancing, none of which are represented. A stronger simulator could make the mechanism more important, but this benchmark cannot prove that.

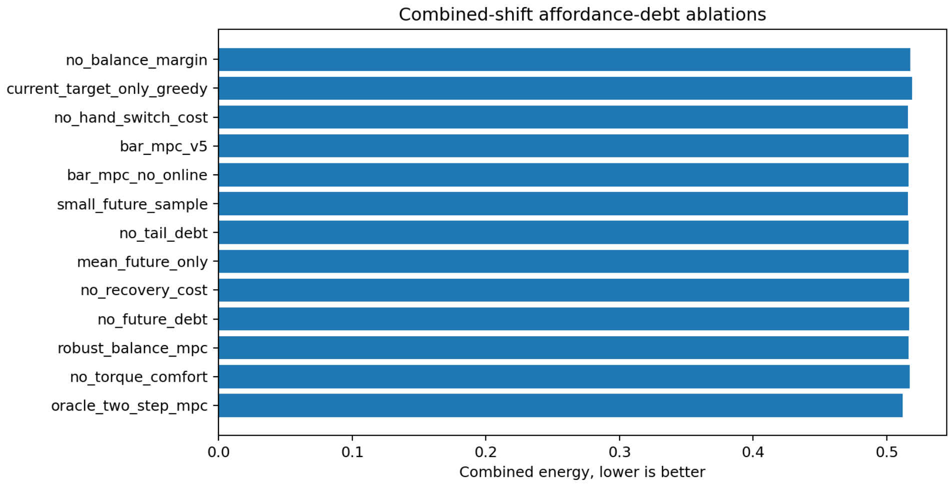

Figure 3: Ablation energy on the hostile ablation splits. The close grouping is not a positive robustness story; it means the proposed debt components are not causally identified by the benchmark.

Table 5: Paired BAR-MPC deltas on hostile splits. Positive success delta and positive energy improvement favor BAR-MPC.

Split	Baseline	Δ success	Energy impr.	Δ margin	First-choice diff.
combined_shift	Greedy reach	0.000	0.000	0.002	21.3%
combined_shift	Comfort MPC	0.000	0.000	0.001	14.1%
combined_shift	Robust balance	0.005	-0.000	0.001	5.7%
combined_shift	Future dist.	0.000	0.000	0.001	10.4%
combined_shift	Linear debt	0.000	-0.000	-0.002	28.6%
combined_shift	BAR no-online	0.000	0.000	0.000	0.0%
combined_shift	Oracle	-0.031	-0.008	0.007	50.5%
opposite_side_sequence	Greedy reach	0.010	0.003	0.005	20.8%
opposite_side_sequence	Comfort MPC	0.005	0.002	0.004	17.2%
opposite_side_sequence	Robust balance	0.000	0.000	0.000	4.2%
opposite_side_sequence	Future dist.	0.005	0.002	0.003	12.5%
opposite_side_sequence	Linear debt	-0.005	-0.001	-0.001	6.2%
opposite_side_sequence	BAR no-online	0.000	0.000	0.000	0.0%
opposite_side_sequence	Oracle	-0.021	-0.004	-0.007	52.6%
support_reversal	Greedy reach	0.010	0.004	0.009	50.5%
support_reversal	Comfort MPC	0.010	0.003	0.007	45.3%
support_reversal	Robust balance	0.000	-0.000	-0.001	9.9%
support_reversal	Future dist.	0.010	0.002	0.007	43.2%
support_reversal	Linear debt	0.005	0.001	0.002	22.9%
support_reversal	BAR no-online	0.000	0.000	0.000	0.0%
support_reversal	Oracle	-0.010	-0.005	0.006	57.8%
high_lateral_payload	Greedy reach	0.000	0.003	0.006	27.1%
high_lateral_payload	Comfort MPC	0.000	0.002	0.005	22.4%
high_lateral_payload	Robust balance	0.000	-0.000	-0.002	8.8%
high_lateral_payload	Future dist.	0.000	0.002	0.005	24.0%
high_lateral_payload	Linear debt	-0.005	0.000	0.001	8.8%
high_lateral_payload	BAR no-online	0.000	0.000	0.000	0.0%
high_lateral_payload	Oracle	-0.005	-0.003	-0.004	45.8%

Failure mode 3: the oracle gap remains. The oracle's advantage in several splits shows that the task is not solved. But an oracle gap alone does not prove the proposed estimator is the right path. The gap could require better dynamics, a richer action set, a learned recovery model, or explicit footstep planning rather than a scalar debt term.

Failure mode 4: no-online equals online. The optional online calibration was not retained because development did not justify it. The frozen no-online ablation therefore matches BAR-MPC. This is honestly reported as a limitation, not hidden behind a renamed method.

Table 6: Pre-registered limitations and negative cases.

Case	Observed limitation	Status
dynamic stepping not modeled	support margin is evaluated with a standing support polygon, not footstep replanning	limitation
out-of-distribution acrobatics	finite posture library cannot represent kneeling, stepping, or hand support	limitation
custom MuJoCo benchmark only	evidence supports strong-revise at best without humanoid hardware or public benchmark validation	limitation

14 What Would Be Needed for a Real Submission

A credible next version would need to change the mechanism, not just expand the table count. The most likely path is to replace the scalar debt proxy with a structured future-option set: reachable target regions, required hand, support mode, and recovery maneuver class. A planner could then preserve discrete options rather than minimizing a continuous penalty. A second path is to move from fixed postures to short-horizon contact sequences, including stepping and hand-supported balance. A third path is to learn a calibrated future-failure model from rollouts and use it only after proving that it changes first choices for the right reason.

The important constraint is that any next version must be frozen before final evaluation. A method that is tuned until it wins this benchmark would not solve the problem; it would only create another fragile artifact.

15 Limitations

The benchmark is custom and compact. It is not hardware, not a public humanoid benchmark suite, and not a full loco-manipulation environment. The robot does not step, kneel, brace with the hand, or perform dynamic whole-body recovery. The future distribution is known as a split-level sampling process rather than inferred from perception. The candidate library is finite, and the simulator does not model many real-world contact uncertainties. These limitations would already prevent an ICLR-main acceptance claim. More importantly, the method fails the internal mechanism tests even under this controlled benchmark.

16 Conclusion

Whole-body affordance debt remains a plausible idea, but this implementation does not provide submission-grade evidence. The frozen results show that BAR-MPC improves over weak baselines but does not survive strong baselines, ablations, and paired hostile diagnostics. The terminal decision is KILL-ARCHIVE. The useful contribution is the falsification: it identifies why a superficially attractive mechanism should not be promoted to a main-conference claim without stronger causal separation.

A Derivation Details

A.1 Two-Stage Objective

The ideal first-stage policy for a two-target episode is

$$\pi^\star(x_0, y_1, \mathcal{Y}) = \arg \min_{c_1 \in \mathcal{C}(y_1)} \ell_{\text{first}}(c_1, y_1) + \mathbb{E}_{y_2 \sim \mathcal{Y}} \left[\min_{c_2 \in \mathcal{C}(y_2)} \ell_{\text{second}}(F(x_0, c_1), c_2, y_2) \right]. \quad (12)$$

This expression is expensive if the action set is rich. The benchmark uses a compact candidate library so that the idealized form is still conceptually visible. BAR-MPC replaces the full second-stage loss with the debt proxy $R(c, y)$ and a tail-risk term.

A.2 Why Greedy Can Tie Debt

Let the candidate score be written as

$$J_\lambda(c) = L(c) + \lambda D(c) + R(c), \quad (13)$$

where L is immediate reach cost, D is future debt, and R is local regularization. If $D(c) = aL(c) + bR(c) + \epsilon(c)$ with small residual ϵ, then

$$J_\lambda(c) = (1 + \lambda a)L(c) + (1 + \lambda b)R(c) + \lambda\epsilon(c). \quad (14)$$

For any candidate pair whose baseline score gap is larger than the residual perturbation,

$$|(L(c_i) + R(c_i)) - (L(c_j) + R(c_j))| > |\lambda(\epsilon(c_i) - \epsilon(c_j))|, \quad (15)$$

the ordering is unchanged. This explains why first-choice differences are limited and why ablations can match the full method.

A.3 A Sufficient Condition for Improvement

Debt can help when there exists a candidate pair a, b satisfying:

$$L(a) > L(b), \quad (16)$$

$$\mathbb{E}_y[R(a, y)] < \mathbb{E}_y[R(b, y)], \quad (17)$$

$$L(a) - L(b) < \lambda_D(\mathbb{E}_y[R(b, y)] - \mathbb{E}_y[R(a, y)]) - (R_{\text{local}}(a) - R_{\text{local}}(b)). \quad (18)$$

The final inequality says the future benefit must be large enough to pay for the current inconvenience. The hostile splits were designed to create such cases, but the frozen evidence shows they are too rare or too weak for the proposed score.

B Implementation Notes

The runner is intentionally small. Each episode samples a split, target pair, future samples, and candidate postures. Candidate first-stage outcomes are simulated once and cached. Second-stage outcomes are computed from the chosen first-stage state. The same prepared episode is shared across methods, so pairwise comparisons are meaningful. Main results and ablations are deterministic given seeds.

The implementation uses MuJoCo through Python, NumPy, SciPy for paired energy tests, and Matplotlib with the non-interactive Agg backend. It writes all final artifacts under `results/`, `figures/`, `docs/`, and `paper/generated/`. The canonical PDF is copied only to `C:/Users/wangz/Downloads/65.pdf`.

C Frozen Command

The final command is:

```
python src\run_experiment.py --seeds 8 --episodes 24
  --splits nominal narrow_support high_reach lateral_reach weak_actuation
  payload_shift combined_shift opposite_side_sequence support_reversal
  high_lateral_payload
  --ablation-splits combined_shift opposite_side_sequence
  --workers 4 --chunksize 1 --results-dir results --figures-dir figures
```

D Reproducibility Checklist

- Code compiles with `python -m py_compile src/run_experiment.py`.
- Main raw rows: 21,120.
- Ablation raw rows: 4,992.
- Main split-method summaries: 110.
- Paired comparison rows: 100.
- Ablation summary rows: 26.
- Figures are regenerated from CSVs.
- Tables in this paper are generated by `scripts/render_submission_assets.py`.
- The validator checks row counts, figure existence, citation-box setup, PDF page count, and absence of a Desktop PDF.

E Full Split-Method Metrics

Table 7: Complete frozen split-method metric table used for the terminal decision.

Split	Method	Episodes	Seq.	Energy	Margin	Bal. fail
combined_shift	Random	192	0.589	0.407	0.076	0.000
combined_shift	Arm only	192	0.698	0.328	0.083	0.000
combined_shift	Greedy reach	192	0.781	0.303	0.072	0.000
combined_shift	Comfort MPC	192	0.781	0.303	0.073	0.000
combined_shift	Robust balance	192	0.776	0.302	0.073	0.000
combined_shift	Current-only	192	0.781	0.303	0.072	0.000
combined_shift	Future dist.	192	0.781	0.303	0.073	0.000
combined_shift	Linear debt	192	0.781	0.302	0.075	0.000
combined_shift	BAR-MPC	192	0.781	0.302	0.074	0.000
combined_shift	BAR no-online	192	0.781	0.302	0.074	0.000
combined_shift	Oracle	192	0.812	0.295	0.067	0.000
high_lateral_payload	Random	192	0.068	0.653	0.057	0.000
high_lateral_payload	Arm only	192	0.073	0.569	0.068	0.000
high_lateral_payload	Greedy reach	192	0.104	0.561	0.037	0.000
high_lateral_payload	Comfort MPC	192	0.104	0.560	0.038	0.000
high_lateral_payload	Robust balance	192	0.104	0.558	0.044	0.000
high_lateral_payload	Current-only	192	0.104	0.561	0.037	0.000
high_lateral_payload	Future dist.	192	0.104	0.560	0.038	0.000
high_lateral_payload	Linear debt	192	0.109	0.559	0.041	0.000
high_lateral_payload	BAR-MPC	192	0.104	0.559	0.043	0.000
high_lateral_payload	BAR no-online	192	0.104	0.559	0.043	0.000
high_lateral_payload	Oracle	192	0.109	0.556	0.047	0.000
high_reach	Random	192	0.703	0.327	0.132	0.000
high_reach	Arm only	192	0.974	0.244	0.137	0.000
high_reach	Greedy reach	192	1.000	0.210	0.138	0.000
high_reach	Comfort MPC	192	1.000	0.210	0.138	0.000
high_reach	Robust balance	192	1.000	0.209	0.138	0.000
high_reach	Current-only	192	1.000	0.210	0.138	0.000
high_reach	Future dist.	192	1.000	0.210	0.138	0.000
high_reach	Linear debt	192	1.000	0.211	0.139	0.000
high_reach	BAR-MPC	192	1.000	0.210	0.138	0.000
high_reach	BAR no-online	192	1.000	0.210	0.138	0.000
high_reach	Oracle	192	1.000	0.207	0.135	0.000
lateral_reach	Random	192	0.281	0.525	0.122	0.000
lateral_reach	Arm only	192	0.354	0.459	0.127	0.000

Split	Method	Episodes	Seq.	Energy	Margin	Bal. fail
lateral_reach	Greedy reach	192	0.453	0.429	0.128	0.000
lateral_reach	Comfort MPC	192	0.453	0.429	0.128	0.000
lateral_reach	Robust balance	192	0.453	0.429	0.128	0.000
lateral_reach	Current-only	192	0.453	0.429	0.128	0.000
lateral_reach	Future dist.	192	0.448	0.429	0.129	0.000
lateral_reach	Linear debt	192	0.448	0.430	0.128	0.000
lateral_reach	BAR-MPC	192	0.448	0.429	0.129	0.000
lateral_reach	BAR no-online	192	0.448	0.429	0.129	0.000
lateral_reach	Oracle	192	0.464	0.426	0.124	0.000
narrow_support	Random	192	0.406	0.460	0.081	0.000
narrow_support	Arm only	192	0.505	0.405	0.085	0.000
narrow_support	Greedy reach	192	0.578	0.385	0.081	0.000
narrow_support	Comfort MPC	192	0.573	0.384	0.080	0.000
narrow_support	Robust balance	192	0.573	0.384	0.081	0.000
narrow_support	Current-only	192	0.578	0.385	0.081	0.000
narrow_support	Future dist.	192	0.573	0.385	0.081	0.000
narrow_support	Linear debt	192	0.568	0.386	0.082	0.000
narrow_support	BAR-MPC	192	0.573	0.384	0.081	0.000
narrow_support	BAR no-online	192	0.573	0.384	0.081	0.000
narrow_support	Oracle	192	0.573	0.383	0.078	0.000
nominal	Random	192	0.312	0.479	0.144	0.000
nominal	Arm only	192	0.500	0.412	0.148	0.000
nominal	Greedy reach	192	0.604	0.389	0.151	0.000
nominal	Comfort MPC	192	0.599	0.388	0.151	0.000
nominal	Robust balance	192	0.599	0.388	0.151	0.000
nominal	Current-only	192	0.604	0.389	0.151	0.000
nominal	Future dist.	192	0.599	0.388	0.152	0.000
nominal	Linear debt	192	0.589	0.390	0.151	0.000
nominal	BAR-MPC	192	0.599	0.388	0.152	0.000
nominal	BAR no-online	192	0.599	0.388	0.152	0.000
nominal	Oracle	192	0.604	0.387	0.150	0.000
opposite_side_sequence	Random	192	0.120	0.626	0.064	0.000
opposite_side_sequence	Arm only	192	0.177	0.527	0.075	0.000
opposite_side_sequence	Greedy reach	192	0.135	0.519	0.056	0.000
opposite_side_sequence	Comfort MPC	192	0.141	0.518	0.057	0.000
opposite_side_sequence	Robust balance	192	0.146	0.516	0.060	0.000
opposite_side_sequence	Current-only	192	0.135	0.519	0.056	0.000
opposite_side_sequence	Future dist.	192	0.141	0.517	0.057	0.000
opposite_side_sequence	Linear debt	192	0.151	0.515	0.061	0.000
opposite_side_sequence	BAR-MPC	192	0.146	0.516	0.060	0.000
opposite_side_sequence	BAR no-online	192	0.146	0.516	0.060	0.000
opposite_side_sequence	Oracle	192	0.167	0.512	0.068	0.000
payload_shift	Random	192	0.354	0.456	0.122	0.000
payload_shift	Arm only	192	0.484	0.387	0.126	0.000
payload_shift	Greedy reach	192	0.536	0.368	0.126	0.000
payload_shift	Comfort MPC	192	0.536	0.368	0.127	0.000
payload_shift	Robust balance	192	0.531	0.368	0.126	0.000
payload_shift	Current-only	192	0.536	0.368	0.126	0.000
payload_shift	Future dist.	192	0.536	0.368	0.127	0.000
payload_shift	Linear debt	192	0.531	0.369	0.127	0.000
payload_shift	BAR-MPC	192	0.536	0.368	0.127	0.000
payload_shift	BAR no-online	192	0.536	0.368	0.127	0.000
payload_shift	Oracle	192	0.542	0.365	0.125	0.000
support_reversal	Random	192	0.146	0.608	0.059	0.000
support_reversal	Arm only	192	0.198	0.515	0.068	0.000
support_reversal	Greedy reach	192	0.167	0.493	0.052	0.000
support_reversal	Comfort MPC	192	0.167	0.492	0.053	0.000

Split	Method	Episodes	Seq.	Energy	Margin	Bal. fail
support_reversal	Robust balance	192	0.177	0.489	0.062	0.000
support_reversal	Current-only	192	0.167	0.493	0.052	0.000
support_reversal	Future dist.	192	0.167	0.491	0.054	0.000
support_reversal	Linear debt	192	0.172	0.489	0.058	0.000
support_reversal	BAR-MPC	192	0.177	0.489	0.060	0.000
support_reversal	BAR no-online	192	0.177	0.489	0.060	0.000
support_reversal	Oracle	192	0.188	0.484	0.054	0.000
weak_actuation	Random	192	0.406	0.432	0.126	0.000
weak_actuation	Arm only	192	0.547	0.372	0.133	0.000
weak_actuation	Greedy reach	192	0.609	0.357	0.135	0.000
weak_actuation	Comfort MPC	192	0.609	0.356	0.135	0.000
weak_actuation	Robust balance	192	0.609	0.356	0.135	0.000
weak_actuation	Current-only	192	0.609	0.357	0.135	0.000
weak_actuation	Future dist.	192	0.599	0.356	0.135	0.000
weak_actuation	Linear debt	192	0.589	0.358	0.135	0.000
weak_actuation	BAR-MPC	192	0.599	0.356	0.136	0.000
weak_actuation	BAR no-online	192	0.599	0.356	0.136	0.000
weak_actuation	Oracle	192	0.594	0.352	0.134	0.000

F Full Paired Comparisons

Table 8: Complete paired BAR-MPC comparison table.

Split	Baseline	Pairs	Δ succ.	Energy impr.	Δ margin	Diff.
combined_shift	Random	192	0.193	0.105	-0.003	92.7%
combined_shift	Arm only	192	0.083	0.025	-0.009	99.0%
combined_shift	Greedy reach	192	0.000	0.000	0.002	21.3%
combined_shift	Comfort MPC	192	0.000	0.000	0.001	14.1%
combined_shift	Robust balance	192	0.005	-0.000	0.001	5.7%
combined_shift	Current-only	192	0.000	0.000	0.002	21.3%
combined_shift	Future dist.	192	0.000	0.000	0.001	10.4%
combined_shift	Linear debt	192	0.000	-0.000	-0.002	28.6%
combined_shift	BAR no-online	192	0.000	0.000	0.000	0.0%
combined_shift	Oracle	192	-0.031	-0.008	0.007	50.5%
high_lateral_payload	Random	192	0.036	0.095	-0.014	95.3%
high_lateral_payload	Arm only	192	0.031	0.011	-0.025	100.0%
high_lateral_payload	Greedy reach	192	0.000	0.003	0.006	27.1%
high_lateral_payload	Comfort MPC	192	0.000	0.002	0.005	22.4%
high_lateral_payload	Robust balance	192	0.000	-0.000	-0.002	8.8%
high_lateral_payload	Current-only	192	0.000	0.003	0.006	27.1%
high_lateral_payload	Future dist.	192	0.000	0.002	0.005	24.0%
high_lateral_payload	Linear debt	192	-0.005	0.000	0.001	8.8%
high_lateral_payload	BAR no-online	192	0.000	0.000	0.000	0.0%
high_lateral_payload	Oracle	192	-0.005	-0.003	-0.004	45.8%
high_reach	Random	192	0.297	0.118	0.007	95.8%
high_reach	Arm only	192	0.026	0.034	0.001	95.8%
high_reach	Greedy reach	192	0.000	0.000	0.001	19.8%
high_reach	Comfort MPC	192	0.000	-0.000	0.000	8.3%
high_reach	Robust balance	192	0.000	-0.000	0.000	8.8%
high_reach	Current-only	192	0.000	0.000	0.001	19.8%
high_reach	Future dist.	192	0.000	-0.000	-0.000	2.6%
high_reach	Linear debt	192	0.000	0.001	-0.001	13.0%
high_reach	BAR no-online	192	0.000	0.000	0.000	0.0%
high_reach	Oracle	192	0.000	-0.003	0.003	29.2%
lateral_reach	Random	192	0.167	0.096	0.006	95.3%
lateral_reach	Arm only	192	0.094	0.030	0.001	94.8%
lateral_reach	Greedy reach	192	-0.005	0.000	0.001	20.3%
lateral_reach	Comfort MPC	192	-0.005	0.000	0.001	13.0%
lateral_reach	Robust balance	192	-0.005	0.000	0.000	13.5%
lateral_reach	Current-only	192	-0.005	0.000	0.001	20.3%
lateral_reach	Future dist.	192	0.000	0.000	0.000	0.0%
lateral_reach	Linear debt	192	0.000	0.002	0.000	26.6%
lateral_reach	BAR no-online	192	0.000	0.000	0.000	0.0%
lateral_reach	Oracle	192	-0.016	-0.003	0.005	35.9%
narrow_support	Random	192	0.167	0.075	-0.000	93.8%
narrow_support	Arm only	192	0.068	0.021	-0.004	93.8%
narrow_support	Greedy reach	192	-0.005	0.000	-0.000	20.3%
narrow_support	Comfort MPC	192	0.000	-0.000	0.000	9.4%
narrow_support	Robust balance	192	0.000	-0.000	-0.000	10.4%
narrow_support	Current-only	192	-0.005	0.000	-0.000	20.3%
narrow_support	Future dist.	192	0.000	0.000	-0.000	2.1%
narrow_support	Linear debt	192	0.005	0.001	-0.001	26.0%
narrow_support	BAR no-online	192	0.000	0.000	0.000	0.0%
narrow_support	Oracle	192	0.000	-0.002	0.003	30.7%
nominal	Random	192	0.286	0.091	0.008	94.8%
nominal	Arm only	192	0.099	0.024	0.004	93.2%
nominal	Greedy reach	192	-0.005	0.000	0.001	24.5%

Split	Baseline	Pairs	Δ succ.	Energy impr.	Δ margin	Diff.
nominal	Comfort MPC	192	0.000	-0.000	0.000	8.8%
nominal	Robust balance	192	0.000	-0.000	0.000	9.9%
nominal	Current-only	192	-0.005	0.000	0.001	24.5%
nominal	Future dist.	192	0.000	-0.000	-0.000	0.5%
nominal	Linear debt	192	0.010	0.002	0.001	34.9%
nominal	BAR no-online	192	0.000	0.000	0.000	0.0%
nominal	Oracle	192	-0.005	-0.001	0.002	29.2%
opposite_side_sequence	Random	192	0.026	0.110	-0.004	91.1%
opposite_side_sequence	Arm only	192	-0.031	0.011	-0.015	100.0%
opposite_side_sequence	Greedy reach	192	0.010	0.003	0.005	20.8%
opposite_side_sequence	Comfort MPC	192	0.005	0.002	0.004	17.2%
opposite_side_sequence	Robust balance	192	0.000	0.000	0.000	4.2%
opposite_side_sequence	Current-only	192	0.010	0.003	0.005	20.8%
opposite_side_sequence	Future dist.	192	0.005	0.002	0.003	12.5%
opposite_side_sequence	Linear debt	192	-0.005	-0.001	-0.001	6.2%
opposite_side_sequence	BAR no-online	192	0.000	0.000	0.000	0.0%
opposite_side_sequence	Oracle	192	-0.021	-0.004	-0.007	52.6%
payload_shift	Random	192	0.182	0.089	0.005	92.7%
payload_shift	Arm only	192	0.052	0.019	0.002	89.1%
payload_shift	Greedy reach	192	0.000	0.001	0.001	24.5%
payload_shift	Comfort MPC	192	0.000	-0.000	0.001	9.9%
payload_shift	Robust balance	192	0.005	0.000	0.001	10.9%
payload_shift	Current-only	192	0.000	0.001	0.001	24.5%
payload_shift	Future dist.	192	0.000	0.000	0.000	1.0%
payload_shift	Linear debt	192	0.005	0.001	-0.000	29.7%
payload_shift	BAR no-online	192	0.000	0.000	0.000	0.0%
payload_shift	Oracle	192	-0.005	-0.003	0.002	41.7%
support_reversal	Random	192	0.031	0.119	0.001	94.3%
support_reversal	Arm only	192	-0.021	0.026	-0.007	100.0%
support_reversal	Greedy reach	192	0.010	0.004	0.009	50.5%
support_reversal	Comfort MPC	192	0.010	0.003	0.007	45.3%
support_reversal	Robust balance	192	0.000	-0.000	-0.001	9.9%
support_reversal	Current-only	192	0.010	0.004	0.009	50.5%
support_reversal	Future dist.	192	0.010	0.002	0.007	43.2%
support_reversal	Linear debt	192	0.005	0.001	0.002	22.9%
support_reversal	BAR no-online	192	0.000	0.000	0.000	0.0%
support_reversal	Oracle	192	-0.010	-0.005	0.006	57.8%
weak_actuation	Random	192	0.193	0.076	0.009	93.8%
weak_actuation	Arm only	192	0.052	0.016	0.003	94.8%
weak_actuation	Greedy reach	192	-0.010	0.000	0.000	16.2%
weak_actuation	Comfort MPC	192	-0.010	-0.000	0.000	10.9%
weak_actuation	Robust balance	192	-0.010	-0.000	0.000	12.0%
weak_actuation	Current-only	192	-0.010	0.000	0.000	16.2%
weak_actuation	Future dist.	192	0.000	0.000	0.000	1.6%
weak_actuation	Linear debt	192	0.010	0.002	0.001	33.9%
weak_actuation	BAR no-online	192	0.000	0.000	0.000	0.0%
weak_actuation	Oracle	192	0.005	-0.004	0.001	43.2%

G SEED-LEVEL ROBUSTNESS

Aggregate means can hide split-seed fragility. Table 9 therefore reports seed-level summaries for the strongest baselines, BAR-MPC, and the oracle. This table is intentionally included in the paper rather than left only in CSV form: hostile review should be able to see whether a conclusion is carried by one lucky seed or appears across the frozen seed grid. The pattern reinforces the terminal decision. Some seeds favor BAR-MPC on hostile splits, but the advantage does not persist strongly enough to survive comparison against robust balance, future-distribution MPC, learned linear debt, and oracle references.

Table 9: Seed-level robustness table for selected strong baselines, BAR-MPC, and the oracle.

Split	Method	Seed	Episodes	Seq.	Energy	Margin
combined_shift	Greedy reach	0	24	0.667	0.319	0.072
combined_shift	Robust balance	0	24	0.667	0.319	0.072
combined_shift	Future dist.	0	24	0.667	0.319	0.072
combined_shift	Linear debt	0	24	0.667	0.319	0.073
combined_shift	BAR-MPC	0	24	0.667	0.319	0.072
combined_shift	Oracle	0	24	0.750	0.314	0.068
combined_shift	Greedy reach	1	24	0.750	0.309	0.075
combined_shift	Robust balance	1	24	0.750	0.308	0.076
combined_shift	Future dist.	1	24	0.750	0.309	0.076
combined_shift	Linear debt	1	24	0.750	0.306	0.077
combined_shift	BAR-MPC	1	24	0.750	0.309	0.077
combined_shift	Oracle	1	24	0.792	0.303	0.073
combined_shift	Greedy reach	2	24	0.667	0.300	0.071
combined_shift	Robust balance	2	24	0.667	0.300	0.074
combined_shift	Future dist.	2	24	0.667	0.300	0.073
combined_shift	Linear debt	2	24	0.708	0.304	0.078
combined_shift	BAR-MPC	2	24	0.667	0.300	0.074
combined_shift	Oracle	2	24	0.792	0.296	0.074
combined_shift	Greedy reach	3	24	0.833	0.294	0.068
combined_shift	Robust balance	3	24	0.833	0.292	0.069
combined_shift	Future dist.	3	24	0.833	0.293	0.070
combined_shift	Linear debt	3	24	0.875	0.292	0.070
combined_shift	BAR-MPC	3	24	0.875	0.292	0.071
combined_shift	Oracle	3	24	0.875	0.284	0.062
combined_shift	Greedy reach	4	24	0.917	0.285	0.075
combined_shift	Robust balance	4	24	0.917	0.283	0.076
combined_shift	Future dist.	4	24	0.917	0.285	0.078
combined_shift	Linear debt	4	24	0.875	0.288	0.081
combined_shift	BAR-MPC	4	24	0.917	0.285	0.078
combined_shift	Oracle	4	24	0.875	0.276	0.069
combined_shift	Greedy reach	5	24	0.750	0.312	0.058
combined_shift	Robust balance	5	24	0.708	0.314	0.061
combined_shift	Future dist.	5	24	0.750	0.314	0.060
combined_shift	Linear debt	5	24	0.708	0.315	0.062
combined_shift	BAR-MPC	5	24	0.708	0.314	0.061
combined_shift	Oracle	5	24	0.750	0.303	0.048
combined_shift	Greedy reach	6	24	0.833	0.277	0.078
combined_shift	Robust balance	6	24	0.833	0.274	0.079
combined_shift	Future dist.	6	24	0.833	0.275	0.079
combined_shift	Linear debt	6	24	0.833	0.272	0.079
combined_shift	BAR-MPC	6	24	0.833	0.274	0.079
combined_shift	Oracle	6	24	0.875	0.268	0.073
combined_shift	Greedy reach	7	24	0.833	0.327	0.076
combined_shift	Robust balance	7	24	0.833	0.326	0.077

Split	Method	Seed	Episodes	Seq.	Energy	Margin
combined_shift	Future dist.	7	24	0.833	0.327	0.077
combined_shift	Linear debt	7	24	0.833	0.323	0.081
combined_shift	BAR-MPC	7	24	0.833	0.326	0.078
combined_shift	Oracle	7	24	0.792	0.316	0.067
high_lateral_payload	Greedy reach	0	24	0.042	0.596	0.037
high_lateral_payload	Robust balance	0	24	0.042	0.593	0.043
high_lateral_payload	Future dist.	0	24	0.042	0.596	0.038
high_lateral_payload	Linear debt	0	24	0.042	0.594	0.041
high_lateral_payload	BAR-MPC	0	24	0.042	0.593	0.043
high_lateral_payload	Oracle	0	24	0.042	0.591	0.048
high_lateral_payload	Greedy reach	1	24	0.250	0.552	0.039
high_lateral_payload	Robust balance	1	24	0.250	0.550	0.044
high_lateral_payload	Future dist.	1	24	0.250	0.552	0.039
high_lateral_payload	Linear debt	1	24	0.250	0.551	0.042
high_lateral_payload	BAR-MPC	1	24	0.250	0.551	0.043
high_lateral_payload	Oracle	1	24	0.208	0.547	0.054
high_lateral_payload	Greedy reach	2	24	0.083	0.531	0.035
high_lateral_payload	Robust balance	2	24	0.083	0.528	0.044
high_lateral_payload	Future dist.	2	24	0.083	0.530	0.036
high_lateral_payload	Linear debt	2	24	0.125	0.527	0.042
high_lateral_payload	BAR-MPC	2	24	0.083	0.528	0.042
high_lateral_payload	Oracle	2	24	0.167	0.525	0.043
high_lateral_payload	Greedy reach	3	24	0.083	0.561	0.035
high_lateral_payload	Robust balance	3	24	0.125	0.559	0.043
high_lateral_payload	Future dist.	3	24	0.083	0.560	0.037
high_lateral_payload	Linear debt	3	24	0.125	0.559	0.039
high_lateral_payload	BAR-MPC	3	24	0.125	0.559	0.042
high_lateral_payload	Oracle	3	24	0.125	0.557	0.044
high_lateral_payload	Greedy reach	4	24	0.042	0.552	0.038
high_lateral_payload	Robust balance	4	24	0.042	0.548	0.046
high_lateral_payload	Future dist.	4	24	0.042	0.551	0.039
high_lateral_payload	Linear debt	4	24	0.042	0.549	0.043
high_lateral_payload	BAR-MPC	4	24	0.042	0.548	0.045
high_lateral_payload	Oracle	4	24	0.042	0.545	0.048
high_lateral_payload	Greedy reach	5	24	0.125	0.549	0.038
high_lateral_payload	Robust balance	5	24	0.125	0.546	0.046
high_lateral_payload	Future dist.	5	24	0.125	0.549	0.038
high_lateral_payload	Linear debt	5	24	0.125	0.547	0.042
high_lateral_payload	BAR-MPC	5	24	0.125	0.547	0.043
high_lateral_payload	Oracle	5	24	0.125	0.543	0.049
high_lateral_payload	Greedy reach	6	24	0.125	0.557	0.037
high_lateral_payload	Robust balance	6	24	0.125	0.556	0.047
high_lateral_payload	Future dist.	6	24	0.125	0.557	0.038
high_lateral_payload	Linear debt	6	24	0.125	0.555	0.043
high_lateral_payload	BAR-MPC	6	24	0.125	0.555	0.044
high_lateral_payload	Oracle	6	24	0.125	0.553	0.045
high_lateral_payload	Greedy reach	7	24	0.083	0.589	0.035
high_lateral_payload	Robust balance	7	24	0.042	0.587	0.041
high_lateral_payload	Future dist.	7	24	0.083	0.589	0.036
high_lateral_payload	Linear debt	7	24	0.042	0.587	0.039
high_lateral_payload	BAR-MPC	7	24	0.042	0.587	0.040
high_lateral_payload	Oracle	7	24	0.042	0.586	0.043
high_reach	Greedy reach	0	24	1.000	0.191	0.134
high_reach	Robust balance	0	24	1.000	0.192	0.134
high_reach	Future dist.	0	24	1.000	0.192	0.135
high_reach	Linear debt	0	24	1.000	0.194	0.137
high_reach	BAR-MPC	0	24	1.000	0.192	0.135

Split	Method	Seed	Episodes	Seq.	Energy	Margin
high_reach	Oracle	0	24	1.000	0.189	0.130
high_reach	Greedy reach	1	24	1.000	0.221	0.129
high_reach	Robust balance	1	24	1.000	0.221	0.131
high_reach	Future dist.	1	24	1.000	0.222	0.132
high_reach	Linear debt	1	24	1.000	0.223	0.133
high_reach	BAR-MPC	1	24	1.000	0.222	0.132
high_reach	Oracle	1	24	1.000	0.219	0.127
high_reach	Greedy reach	2	24	1.000	0.194	0.140
high_reach	Robust balance	2	24	1.000	0.194	0.139
high_reach	Future dist.	2	24	1.000	0.195	0.139
high_reach	Linear debt	2	24	1.000	0.195	0.139
high_reach	BAR-MPC	2	24	1.000	0.195	0.139
high_reach	Oracle	2	24	1.000	0.192	0.139
high_reach	Greedy reach	3	24	1.000	0.193	0.144
high_reach	Robust balance	3	24	1.000	0.191	0.145
high_reach	Future dist.	3	24	1.000	0.191	0.144
high_reach	Linear debt	3	24	1.000	0.194	0.144
high_reach	BAR-MPC	3	24	1.000	0.191	0.143
high_reach	Oracle	3	24	1.000	0.188	0.140
high_reach	Greedy reach	4	24	1.000	0.231	0.142
high_reach	Robust balance	4	24	1.000	0.230	0.141
high_reach	Future dist.	4	24	1.000	0.230	0.141
high_reach	Linear debt	4	24	1.000	0.230	0.141
high_reach	BAR-MPC	4	24	1.000	0.230	0.141
high_reach	Oracle	4	24	1.000	0.229	0.136
high_reach	Greedy reach	5	24	1.000	0.216	0.142
high_reach	Robust balance	5	24	1.000	0.216	0.143
high_reach	Future dist.	5	24	1.000	0.217	0.143
high_reach	Linear debt	5	24	1.000	0.217	0.142
high_reach	BAR-MPC	5	24	1.000	0.217	0.143
high_reach	Oracle	5	24	1.000	0.216	0.142
high_reach	Greedy reach	6	24	1.000	0.195	0.137
high_reach	Robust balance	6	24	1.000	0.195	0.137
high_reach	Future dist.	6	24	1.000	0.195	0.137
high_reach	Linear debt	6	24	1.000	0.197	0.139
high_reach	BAR-MPC	6	24	1.000	0.196	0.138
high_reach	Oracle	6	24	1.000	0.194	0.136
high_reach	Greedy reach	7	24	1.000	0.238	0.136
high_reach	Robust balance	7	24	1.000	0.236	0.135
high_reach	Future dist.	7	24	1.000	0.237	0.137
high_reach	Linear debt	7	24	1.000	0.237	0.137
high_reach	BAR-MPC	7	24	1.000	0.237	0.137
high_reach	Oracle	7	24	1.000	0.232	0.130
lateral_reach	Greedy reach	0	24	0.417	0.427	0.125
lateral_reach	Robust balance	0	24	0.417	0.427	0.126
lateral_reach	Future dist.	0	24	0.417	0.427	0.126
lateral_reach	Linear debt	0	24	0.417	0.431	0.127
lateral_reach	BAR-MPC	0	24	0.417	0.427	0.126
lateral_reach	Oracle	0	24	0.458	0.423	0.118
lateral_reach	Greedy reach	1	24	0.417	0.455	0.125
lateral_reach	Robust balance	1	24	0.417	0.454	0.124
lateral_reach	Future dist.	1	24	0.417	0.453	0.123
lateral_reach	Linear debt	1	24	0.417	0.455	0.123
lateral_reach	BAR-MPC	1	24	0.417	0.453	0.123
lateral_reach	Oracle	1	24	0.417	0.452	0.119
lateral_reach	Greedy reach	2	24	0.583	0.417	0.119
lateral_reach	Robust balance	2	24	0.583	0.414	0.121

Split	Method	Seed	Episodes	Seq.	Energy	Margin
lateral_reach	Future dist.	2	24	0.542	0.414	0.121
lateral_reach	Linear debt	2	24	0.542	0.416	0.120
lateral_reach	BAR-MPC	2	24	0.542	0.414	0.121
lateral_reach	Oracle	2	24	0.583	0.412	0.118
lateral_reach	Greedy reach	3	24	0.417	0.432	0.130
lateral_reach	Robust balance	3	24	0.417	0.432	0.129
lateral_reach	Future dist.	3	24	0.417	0.431	0.132
lateral_reach	Linear debt	3	24	0.417	0.432	0.132
lateral_reach	BAR-MPC	3	24	0.417	0.431	0.132
lateral_reach	Oracle	3	24	0.458	0.427	0.126
lateral_reach	Greedy reach	4	24	0.500	0.422	0.129
lateral_reach	Robust balance	4	24	0.500	0.422	0.129
lateral_reach	Future dist.	4	24	0.500	0.422	0.130
lateral_reach	Linear debt	4	24	0.500	0.423	0.129
lateral_reach	BAR-MPC	4	24	0.500	0.422	0.130
lateral_reach	Oracle	4	24	0.458	0.420	0.124
lateral_reach	Greedy reach	5	24	0.375	0.450	0.133
lateral_reach	Robust balance	5	24	0.375	0.450	0.133
lateral_reach	Future dist.	5	24	0.375	0.450	0.133
lateral_reach	Linear debt	5	24	0.375	0.451	0.132
lateral_reach	BAR-MPC	5	24	0.375	0.450	0.133
lateral_reach	Oracle	5	24	0.417	0.447	0.128
lateral_reach	Greedy reach	6	24	0.458	0.434	0.134
lateral_reach	Robust balance	6	24	0.458	0.435	0.134
lateral_reach	Future dist.	6	24	0.458	0.434	0.135
lateral_reach	Linear debt	6	24	0.458	0.437	0.136
lateral_reach	BAR-MPC	6	24	0.458	0.434	0.135
lateral_reach	Oracle	6	24	0.458	0.432	0.132
lateral_reach	Greedy reach	7	24	0.458	0.396	0.127
lateral_reach	Robust balance	7	24	0.458	0.397	0.128
lateral_reach	Future dist.	7	24	0.458	0.398	0.129
lateral_reach	Linear debt	7	24	0.458	0.398	0.129
lateral_reach	BAR-MPC	7	24	0.458	0.398	0.129
lateral_reach	Oracle	7	24	0.458	0.394	0.126
narrow_support	Greedy reach	0	24	0.542	0.395	0.081
narrow_support	Robust balance	0	24	0.500	0.395	0.080
narrow_support	Future dist.	0	24	0.500	0.396	0.082
narrow_support	Linear debt	0	24	0.458	0.399	0.085
narrow_support	BAR-MPC	0	24	0.500	0.396	0.082
narrow_support	Oracle	0	24	0.500	0.395	0.079
narrow_support	Greedy reach	1	24	0.667	0.361	0.086
narrow_support	Robust balance	1	24	0.667	0.361	0.084
narrow_support	Future dist.	1	24	0.667	0.361	0.081
narrow_support	Linear debt	1	24	0.667	0.361	0.082
narrow_support	BAR-MPC	1	24	0.667	0.361	0.081
narrow_support	Oracle	1	24	0.667	0.359	0.077
narrow_support	Greedy reach	2	24	0.417	0.388	0.077
narrow_support	Robust balance	2	24	0.417	0.388	0.077
narrow_support	Future dist.	2	24	0.417	0.387	0.077
narrow_support	Linear debt	2	24	0.417	0.389	0.079
narrow_support	BAR-MPC	2	24	0.417	0.387	0.077
narrow_support	Oracle	2	24	0.417	0.387	0.076
narrow_support	Greedy reach	3	24	0.625	0.401	0.083
narrow_support	Robust balance	3	24	0.625	0.401	0.084
narrow_support	Future dist.	3	24	0.625	0.401	0.085
narrow_support	Linear debt	3	24	0.625	0.404	0.087
narrow_support	BAR-MPC	3	24	0.625	0.401	0.085

Split	Method	Seed	Episodes	Seq.	Energy	Margin
narrow_support	Oracle	3	24	0.583	0.399	0.082
narrow_support	Greedy reach	4	24	0.458	0.403	0.083
narrow_support	Robust balance	4	24	0.458	0.402	0.083
narrow_support	Future dist.	4	24	0.458	0.402	0.083
narrow_support	Linear debt	4	24	0.458	0.402	0.082
narrow_support	BAR-MPC	4	24	0.458	0.402	0.083
narrow_support	Oracle	4	24	0.458	0.400	0.081
narrow_support	Greedy reach	5	24	0.792	0.366	0.077
narrow_support	Robust balance	5	24	0.792	0.365	0.077
narrow_support	Future dist.	5	24	0.792	0.366	0.078
narrow_support	Linear debt	5	24	0.792	0.366	0.078
narrow_support	BAR-MPC	5	24	0.792	0.366	0.078
narrow_support	Oracle	5	24	0.792	0.363	0.075
narrow_support	Greedy reach	6	24	0.500	0.407	0.081
narrow_support	Robust balance	6	24	0.500	0.406	0.082
narrow_support	Future dist.	6	24	0.500	0.405	0.083
narrow_support	Linear debt	6	24	0.500	0.406	0.083
narrow_support	BAR-MPC	6	24	0.500	0.405	0.082
narrow_support	Oracle	6	24	0.500	0.402	0.077
narrow_support	Greedy reach	7	24	0.625	0.358	0.079
narrow_support	Robust balance	7	24	0.625	0.357	0.078
narrow_support	Future dist.	7	24	0.625	0.358	0.077
narrow_support	Linear debt	7	24	0.625	0.360	0.080
narrow_support	BAR-MPC	7	24	0.625	0.358	0.077
narrow_support	Oracle	7	24	0.667	0.357	0.078
nominal	Greedy reach	0	24	0.542	0.415	0.147
nominal	Robust balance	0	24	0.542	0.414	0.148
nominal	Future dist.	0	24	0.542	0.414	0.148
nominal	Linear debt	0	24	0.542	0.415	0.146
nominal	BAR-MPC	0	24	0.542	0.414	0.148
nominal	Oracle	0	24	0.542	0.412	0.145
nominal	Greedy reach	1	24	0.667	0.378	0.148
nominal	Robust balance	1	24	0.667	0.379	0.149
nominal	Future dist.	1	24	0.667	0.379	0.149
nominal	Linear debt	1	24	0.667	0.380	0.148
nominal	BAR-MPC	1	24	0.667	0.379	0.149
nominal	Oracle	1	24	0.667	0.378	0.149
nominal	Greedy reach	2	24	0.458	0.395	0.155
nominal	Robust balance	2	24	0.417	0.394	0.153
nominal	Future dist.	2	24	0.417	0.394	0.153
nominal	Linear debt	2	24	0.417	0.395	0.154
nominal	BAR-MPC	2	24	0.417	0.394	0.153
nominal	Oracle	2	24	0.458	0.394	0.154
nominal	Greedy reach	3	24	0.542	0.389	0.153
nominal	Robust balance	3	24	0.542	0.389	0.153
nominal	Future dist.	3	24	0.542	0.389	0.153
nominal	Linear debt	3	24	0.500	0.392	0.151
nominal	BAR-MPC	3	24	0.542	0.389	0.153
nominal	Oracle	3	24	0.542	0.389	0.153
nominal	Greedy reach	4	24	0.833	0.336	0.146
nominal	Robust balance	4	24	0.833	0.336	0.146
nominal	Future dist.	4	24	0.833	0.336	0.147
nominal	Linear debt	4	24	0.792	0.339	0.148
nominal	BAR-MPC	4	24	0.833	0.336	0.147
nominal	Oracle	4	24	0.792	0.336	0.144
nominal	Greedy reach	5	24	0.750	0.389	0.153
nominal	Robust balance	5	24	0.750	0.389	0.153

Split	Method	Seed	Episodes	Seq.	Energy	Margin
nominal	Future dist.	5	24	0.750	0.389	0.153
nominal	Linear debt	5	24	0.750	0.392	0.150
nominal	BAR-MPC	5	24	0.750	0.389	0.153
nominal	Oracle	5	24	0.750	0.387	0.150
nominal	Greedy reach	6	24	0.625	0.392	0.153
nominal	Robust balance	6	24	0.625	0.392	0.153
nominal	Future dist.	6	24	0.625	0.393	0.155
nominal	Linear debt	6	24	0.625	0.396	0.154
nominal	BAR-MPC	6	24	0.625	0.393	0.155
nominal	Oracle	6	24	0.625	0.391	0.150
nominal	Greedy reach	7	24	0.417	0.414	0.154
nominal	Robust balance	7	24	0.417	0.412	0.153
nominal	Future dist.	7	24	0.417	0.413	0.154
nominal	Linear debt	7	24	0.417	0.415	0.155
nominal	BAR-MPC	7	24	0.417	0.413	0.154
nominal	Oracle	7	24	0.458	0.412	0.155
opposite_side_sequence	Greedy reach	0	24	0.083	0.502	0.057
opposite_side_sequence	Robust balance	0	24	0.083	0.499	0.062
opposite_side_sequence	Future dist.	0	24	0.083	0.500	0.060
opposite_side_sequence	Linear debt	0	24	0.083	0.499	0.063
opposite_side_sequence	BAR-MPC	0	24	0.083	0.499	0.063
opposite_side_sequence	Oracle	0	24	0.083	0.496	0.067
opposite_side_sequence	Greedy reach	1	24	0.042	0.524	0.059
opposite_side_sequence	Robust balance	1	24	0.042	0.523	0.062
opposite_side_sequence	Future dist.	1	24	0.042	0.524	0.059
opposite_side_sequence	Linear debt	1	24	0.042	0.521	0.064
opposite_side_sequence	BAR-MPC	1	24	0.042	0.522	0.063
opposite_side_sequence	Oracle	1	24	0.042	0.518	0.072
opposite_side_sequence	Greedy reach	2	24	0.125	0.513	0.051
opposite_side_sequence	Robust balance	2	24	0.167	0.511	0.054
opposite_side_sequence	Future dist.	2	24	0.125	0.513	0.051
opposite_side_sequence	Linear debt	2	24	0.167	0.511	0.054
opposite_side_sequence	BAR-MPC	2	24	0.167	0.511	0.054
opposite_side_sequence	Oracle	2	24	0.167	0.508	0.063
opposite_side_sequence	Greedy reach	3	24	0.167	0.529	0.054
opposite_side_sequence	Robust balance	3	24	0.167	0.524	0.061
opposite_side_sequence	Future dist.	3	24	0.167	0.527	0.056
opposite_side_sequence	Linear debt	3	24	0.167	0.523	0.062
opposite_side_sequence	BAR-MPC	3	24	0.167	0.524	0.060
opposite_side_sequence	Oracle	3	24	0.250	0.518	0.069
opposite_side_sequence	Greedy reach	4	24	0.375	0.488	0.053
opposite_side_sequence	Robust balance	4	24	0.375	0.486	0.057
opposite_side_sequence	Future dist.	4	24	0.375	0.487	0.054
opposite_side_sequence	Linear debt	4	24	0.375	0.485	0.059
opposite_side_sequence	BAR-MPC	4	24	0.375	0.486	0.058
opposite_side_sequence	Oracle	4	24	0.375	0.482	0.067
opposite_side_sequence	Greedy reach	5	24	0.083	0.545	0.057
opposite_side_sequence	Robust balance	5	24	0.083	0.544	0.061
opposite_side_sequence	Future dist.	5	24	0.083	0.544	0.061
opposite_side_sequence	Linear debt	5	24	0.125	0.542	0.060
opposite_side_sequence	BAR-MPC	5	24	0.083	0.544	0.061
opposite_side_sequence	Oracle	5	24	0.125	0.537	0.069
opposite_side_sequence	Greedy reach	6	24	0.083	0.540	0.052
opposite_side_sequence	Robust balance	6	24	0.125	0.534	0.060
opposite_side_sequence	Future dist.	6	24	0.125	0.536	0.057
opposite_side_sequence	Linear debt	6	24	0.125	0.533	0.061
opposite_side_sequence	BAR-MPC	6	24	0.125	0.534	0.060

Split	Method	Seed	Episodes	Seq.	Energy	Margin
opposite_side_sequence	Oracle	6	24	0.125	0.530	0.067
opposite_side_sequence	Greedy reach	7	24	0.125	0.510	0.061
opposite_side_sequence	Robust balance	7	24	0.125	0.509	0.062
opposite_side_sequence	Future dist.	7	24	0.125	0.508	0.062
opposite_side_sequence	Linear debt	7	24	0.125	0.508	0.063
opposite_side_sequence	BAR-MPC	7	24	0.125	0.508	0.063
opposite_side_sequence	Oracle	7	24	0.167	0.505	0.068
payload_shift	Greedy reach	0	24	0.458	0.388	0.126
payload_shift	Robust balance	0	24	0.458	0.387	0.126
payload_shift	Future dist.	0	24	0.458	0.387	0.126
payload_shift	Linear debt	0	24	0.458	0.387	0.126
payload_shift	BAR-MPC	0	24	0.458	0.387	0.126
payload_shift	Oracle	0	24	0.458	0.385	0.122
payload_shift	Greedy reach	1	24	0.417	0.399	0.131
payload_shift	Robust balance	1	24	0.417	0.398	0.130
payload_shift	Future dist.	1	24	0.417	0.399	0.132
payload_shift	Linear debt	1	24	0.458	0.400	0.132
payload_shift	BAR-MPC	1	24	0.417	0.399	0.132
payload_shift	Oracle	1	24	0.417	0.397	0.129
payload_shift	Greedy reach	2	24	0.583	0.347	0.119
payload_shift	Robust balance	2	24	0.583	0.346	0.120
payload_shift	Future dist.	2	24	0.625	0.343	0.122
payload_shift	Linear debt	2	24	0.625	0.344	0.122
payload_shift	BAR-MPC	2	24	0.625	0.343	0.122
payload_shift	Oracle	2	24	0.625	0.341	0.121
payload_shift	Greedy reach	3	24	0.667	0.359	0.116
payload_shift	Robust balance	3	24	0.667	0.359	0.117
payload_shift	Future dist.	3	24	0.667	0.359	0.117
payload_shift	Linear debt	3	24	0.583	0.361	0.119
payload_shift	BAR-MPC	3	24	0.667	0.359	0.117
payload_shift	Oracle	3	24	0.667	0.355	0.118
payload_shift	Greedy reach	4	24	0.667	0.363	0.134
payload_shift	Robust balance	4	24	0.625	0.363	0.133
payload_shift	Future dist.	4	24	0.625	0.363	0.132
payload_shift	Linear debt	4	24	0.625	0.362	0.130
payload_shift	BAR-MPC	4	24	0.625	0.363	0.132
payload_shift	Oracle	4	24	0.667	0.359	0.128
payload_shift	Greedy reach	5	24	0.542	0.341	0.126
payload_shift	Robust balance	5	24	0.542	0.340	0.129
payload_shift	Future dist.	5	24	0.542	0.340	0.129
payload_shift	Linear debt	5	24	0.542	0.342	0.129
payload_shift	BAR-MPC	5	24	0.542	0.340	0.129
payload_shift	Oracle	5	24	0.583	0.339	0.131
payload_shift	Greedy reach	6	24	0.500	0.381	0.124
payload_shift	Robust balance	6	24	0.500	0.380	0.124
payload_shift	Future dist.	6	24	0.500	0.380	0.126
payload_shift	Linear debt	6	24	0.500	0.383	0.127
payload_shift	BAR-MPC	6	24	0.500	0.380	0.126
payload_shift	Oracle	6	24	0.458	0.377	0.126
payload_shift	Greedy reach	7	24	0.458	0.369	0.132
payload_shift	Robust balance	7	24	0.458	0.369	0.133
payload_shift	Future dist.	7	24	0.458	0.370	0.133
payload_shift	Linear debt	7	24	0.458	0.371	0.132
payload_shift	BAR-MPC	7	24	0.458	0.369	0.133
payload_shift	Oracle	7	24	0.458	0.367	0.130
support_reversal	Greedy reach	0	24	0.083	0.503	0.052
support_reversal	Robust balance	0	24	0.083	0.499	0.061

Split	Method	Seed	Episodes	Seq.	Energy	Margin
support_reversal	Future dist.	0	24	0.083	0.500	0.055
support_reversal	Linear debt	0	24	0.083	0.499	0.060
support_reversal	BAR-MPC	0	24	0.083	0.500	0.061
support_reversal	Oracle	0	24	0.083	0.494	0.056
support_reversal	Greedy reach	1	24	0.167	0.485	0.050
support_reversal	Robust balance	1	24	0.167	0.481	0.062
support_reversal	Future dist.	1	24	0.167	0.485	0.050
support_reversal	Linear debt	1	24	0.208	0.482	0.058
support_reversal	BAR-MPC	1	24	0.167	0.481	0.061
support_reversal	Oracle	1	24	0.167	0.477	0.056
support_reversal	Greedy reach	2	24	0.167	0.483	0.053
support_reversal	Robust balance	2	24	0.167	0.479	0.063
support_reversal	Future dist.	2	24	0.167	0.482	0.057
support_reversal	Linear debt	2	24	0.167	0.481	0.059
support_reversal	BAR-MPC	2	24	0.167	0.480	0.061
support_reversal	Oracle	2	24	0.167	0.476	0.054
support_reversal	Greedy reach	3	24	0.208	0.491	0.050
support_reversal	Robust balance	3	24	0.167	0.486	0.058
support_reversal	Future dist.	3	24	0.208	0.490	0.050
support_reversal	Linear debt	3	24	0.167	0.490	0.053
support_reversal	BAR-MPC	3	24	0.167	0.489	0.054
support_reversal	Oracle	3	24	0.167	0.482	0.051
support_reversal	Greedy reach	4	24	0.125	0.506	0.050
support_reversal	Robust balance	4	24	0.125	0.502	0.060
support_reversal	Future dist.	4	24	0.125	0.502	0.053
support_reversal	Linear debt	4	24	0.125	0.502	0.058
support_reversal	BAR-MPC	4	24	0.125	0.501	0.060
support_reversal	Oracle	4	24	0.125	0.497	0.053
support_reversal	Greedy reach	5	24	0.292	0.484	0.048
support_reversal	Robust balance	5	24	0.333	0.481	0.056
support_reversal	Future dist.	5	24	0.292	0.483	0.049
support_reversal	Linear debt	5	24	0.333	0.480	0.054
support_reversal	BAR-MPC	5	24	0.333	0.480	0.056
support_reversal	Oracle	5	24	0.417	0.476	0.049
support_reversal	Greedy reach	6	24	0.125	0.498	0.052
support_reversal	Robust balance	6	24	0.167	0.493	0.064
support_reversal	Future dist.	6	24	0.125	0.496	0.053
support_reversal	Linear debt	6	24	0.125	0.493	0.061
support_reversal	BAR-MPC	6	24	0.167	0.492	0.063
support_reversal	Oracle	6	24	0.208	0.487	0.056
support_reversal	Greedy reach	7	24	0.167	0.491	0.060
support_reversal	Robust balance	7	24	0.208	0.488	0.069
support_reversal	Future dist.	7	24	0.167	0.490	0.062
support_reversal	Linear debt	7	24	0.167	0.488	0.064
support_reversal	BAR-MPC	7	24	0.208	0.489	0.067
support_reversal	Oracle	7	24	0.167	0.483	0.057
weak_actuation	Greedy reach	0	24	0.583	0.369	0.133
weak_actuation	Robust balance	0	24	0.583	0.369	0.133
weak_actuation	Future dist.	0	24	0.583	0.370	0.133
weak_actuation	Linear debt	0	24	0.583	0.374	0.130
weak_actuation	BAR-MPC	0	24	0.583	0.370	0.133
weak_actuation	Oracle	0	24	0.542	0.366	0.133
weak_actuation	Greedy reach	1	24	0.708	0.336	0.139
weak_actuation	Robust balance	1	24	0.708	0.336	0.139
weak_actuation	Future dist.	1	24	0.708	0.335	0.139
weak_actuation	Linear debt	1	24	0.708	0.339	0.141
weak_actuation	BAR-MPC	1	24	0.708	0.335	0.139

Split	Method	Seed	Episodes	Seq.	Energy	Margin
weak_actuation	Oracle	1	24	0.708	0.333	0.140
weak_actuation	Greedy reach	2	24	0.583	0.333	0.137
weak_actuation	Robust balance	2	24	0.583	0.333	0.137
weak_actuation	Future dist.	2	24	0.583	0.333	0.137
weak_actuation	Linear debt	2	24	0.542	0.334	0.134
weak_actuation	BAR-MPC	2	24	0.583	0.333	0.137
weak_actuation	Oracle	2	24	0.542	0.329	0.134
weak_actuation	Greedy reach	3	24	0.458	0.389	0.135
weak_actuation	Robust balance	3	24	0.458	0.388	0.134
weak_actuation	Future dist.	3	24	0.417	0.388	0.134
weak_actuation	Linear debt	3	24	0.417	0.391	0.134
weak_actuation	BAR-MPC	3	24	0.417	0.388	0.134
weak_actuation	Oracle	3	24	0.500	0.386	0.132
weak_actuation	Greedy reach	4	24	0.625	0.377	0.135
weak_actuation	Robust balance	4	24	0.625	0.377	0.135
weak_actuation	Future dist.	4	24	0.583	0.377	0.134
weak_actuation	Linear debt	4	24	0.583	0.378	0.135
weak_actuation	BAR-MPC	4	24	0.583	0.377	0.134
weak_actuation	Oracle	4	24	0.583	0.371	0.135
weak_actuation	Greedy reach	5	24	0.500	0.382	0.137
weak_actuation	Robust balance	5	24	0.500	0.383	0.136
weak_actuation	Future dist.	5	24	0.500	0.384	0.137
weak_actuation	Linear debt	5	24	0.542	0.383	0.137
weak_actuation	BAR-MPC	5	24	0.500	0.383	0.137
weak_actuation	Oracle	5	24	0.542	0.377	0.137
weak_actuation	Greedy reach	6	24	0.708	0.319	0.130
weak_actuation	Robust balance	6	24	0.708	0.317	0.132
weak_actuation	Future dist.	6	24	0.708	0.317	0.133
weak_actuation	Linear debt	6	24	0.708	0.318	0.133
weak_actuation	BAR-MPC	6	24	0.708	0.317	0.134
weak_actuation	Oracle	6	24	0.708	0.313	0.134
weak_actuation	Greedy reach	7	24	0.708	0.346	0.136
weak_actuation	Robust balance	7	24	0.708	0.346	0.137
weak_actuation	Future dist.	7	24	0.708	0.346	0.137
weak_actuation	Linear debt	7	24	0.625	0.345	0.134
weak_actuation	BAR-MPC	7	24	0.708	0.346	0.137
weak_actuation	Oracle	7	24	0.625	0.338	0.131

H Hostile Reviewer Questions

Does the benchmark prove affordance debt is impossible? No. It proves that this implementation does not provide enough evidence. The idea may still work with richer contact modes, learned failure prediction, footstep planning, or a better option representation.

Why not submit as a negative result? A negative result can be valuable, but ICLR-main acceptance still requires a sufficiently broad and externally convincing study. This paper has a custom compact simulator and no hardware or public-suite validation. It is submission-grade as an internal archive report, not as a main-conference claim.

Could stronger tuning make BAR-MPC win? Possibly on this benchmark, but post-freeze tuning would be scientifically weak. The purpose of the expanded protocol is to avoid optimizing for pretty results. If a new method is developed, it should receive a new pre-freeze development log and a new frozen evaluation.

Why include the oracle? The oracle is not a fair deployed baseline. It indicates how much room remains if the second-stage target is known. The oracle gap shows that the benchmark contains avoidable failures, but it does not validate BAR-MPC.

Why keep a 25+ page PDF for an archived paper? Because the expanded standard is about evidence quality, not outcome optimism. A paper that fails should still fail with enough detail that the next attempt knows exactly what not to repeat.

I Terminal Decision

The final decision is KILL-ARCHIVE. The evidence supports archiving Paper 65 under the expanded standard. The code, frozen protocol, generated tables, validation script, and final PDF remain useful for future work, but the current mechanism should not be sent to ICLR main as a positive contribution.

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